

Study Material – Introduction to Machine Learning

CO1: Understand the concept of machine learning, types of learning, and applications

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that focuses on the development of systems that can learn from data.

Types of Machine Learning:

1. Supervised Learning: Uses labeled data to train models (e.g., Linear Regression, Decision Trees).
2. Unsupervised Learning: Works with unlabeled data to identify patterns (e.g., Clustering).
3. Semi-supervised Learning: Mix of labeled and unlabeled data.
4. Reinforcement Learning: An agent learns by interacting with its environment and receiving rewards or penalties.

Applications:

- Email spam filtering
- Product recommendations
- Medical diagnosis
- Fraud detection

CO2: Formulate machine learning problems and understand performance evaluation metrics

Formulating ML Problems:

- Identify features (input variables) and labels (target output).
- Decide whether it's a classification or regression problem.

Model Evaluation Metrics:

- Accuracy: $\text{Correct predictions} / \text{total predictions}$.
- Precision: $\text{True positives} / (\text{true positives} + \text{false positives})$.
- Recall: $\text{True positives} / (\text{true positives} + \text{false negatives})$.
- F1 Score: Harmonic mean of precision and recall.
- Confusion Matrix: Summary of prediction results.
- ROC-AUC: Measures performance across all classification thresholds.

Cross-validation helps evaluate model generalization by splitting data into multiple training/testing sets.

CO3: Understand the basic learning algorithms – regression, classification, clustering

Regression:

- Linear Regression: Models relationship between a dependent variable and one or more independent variables.
- Logistic Regression: Used for binary classification problems.

Classification:

- k-Nearest Neighbors (k-NN): Classifies based on the majority label of k closest points.
- Decision Trees: Splits data based on feature values.
- Naive Bayes: Probabilistic classifier based on Bayes' Theorem.

Clustering:

- k-Means Clustering: Partitions data into k groups by minimizing intra-cluster variance.
- Hierarchical Clustering: Builds a tree of clusters.

CO4: Learn feature engineering, model selection, and tuning techniques

Feature Engineering:

- Data Cleaning: Handle missing values, outliers.
- Normalization: Scale features to a similar range.
- Feature Selection: Choose relevant features.
- Dimensionality Reduction: Techniques like PCA reduce the number of input variables.

Model Selection:

- Compare multiple models using metrics.
- Use validation set or cross-validation.

Model Tuning:

- Hyperparameters: Not learned from data (e.g., learning rate, number of trees).
- Grid Search: Try all combinations of specified hyperparameters.

CO5: Explore modern machine learning topics: ensemble methods, neural networks

Ensemble Methods:

- Bagging: Combine predictions from multiple models trained on random subsets (e.g., Random Forest).
- Boosting: Combine weak learners sequentially to build a strong model (e.g., AdaBoost, Gradient Boosting).

Neural Networks:

- Perceptron: Basic neural unit with weighted inputs and activation function.
- Activation Functions: ReLU, Sigmoid, Tanh – decide neuron output.
- Backpropagation: Algorithm to update weights by minimizing loss.

Deep Learning:

- Convolutional Neural Networks (CNN): Specialized for image data.
- Recurrent Neural Networks (RNN): Handle sequential data like text or time series.